

Team Astra: EMIT scene handover

From Marwa (Stage 2, find the data) to Asma | Arab Youth Space Hackathon, 813 Challenge | 9 October 2026

Status

EMIT scene found and checked against Asma's exact box: **West 6.903, South 33.422, East 7.019, North 33.523** (10.8 km x 11.2 km, centre 33.4725 N, 6.961 E). All four corners are inside the scene, the nearest about 24 km from its edge. The MASK file was opened: **no cloud over the box** (cloud, cirrus and dilated-cloud flags 0.0%). The catalogue value of 100% cloud cover is wrong for this area. Aerosol optical depth is high (about 0.71), which suggests dust haze. **One point is unclear:** the Aggregate flag is 100% over the box, see the table.

The scene

Product	EMIT L2A Estimated Surface Reflectance and Uncertainty and Masks 60 m V001 (EMITL2ARFL)
Granule ID	EMIT_L2A_RFL_001_20230127T133045_2302709_042
Date / time	2023-01-27, 13:30:45 to 13:30:56 UTC (daytime)
Footprint	Tilted rectangle, corners (lat, lon): 33.94, 6.98 / 33.32, 6.23 / 32.71, 6.74 / 33.33, 7.49. Contains the El Oued point 33.37, 6.86.
Cloud (MASK file)	Over the box: Cloud 0.0%, Cirrus 0.0%, Dilated cloud 0.0% (about 29,000 pixels of 60 m). Whole scene: cloud flag 0.1%. The catalogue says 100%, which is wrong.
Dust (MASK file)	AOD at 550 nm over the box: mean 0.71 (0.66 to 0.77). High value, consistent with the pale Sentinel-2 images of 2023-01-18, 01-23 and 01-28.
Unclear flags	Aggregate, Water and Spacecraft flags are all 100% over the box. Water and Spacecraft are identical everywhere and set on 97.5% of the scene, which does not look like real water or spacecraft over desert. So the Aggregate flag cannot be read as cloud or as dust. To be checked in the LP DAAC user guide (EMITL2ARFL) before relying on it.
Link	Open this scene in NASA Earthdata Search

Files (3, about 3.5 GB in total)

MASK (44 MB): already downloaded and read. **RFL** (reflectance, about 3 GB): the data to use, not downloaded yet.

RFLUNCERT (uncertainty): not needed at first.

To download: create a free account at urs.earthdata.nasa.gov, open the link above, log in (EDL), and click the download icon next to the granule. For big files use Earthdata Download or the Download Script tab, not the browser alone.

Sentinel-2 dates (Copernicus Browser, L2A, 2022-11-04 to 2023-03-29, 5-day step)

Screened by eye on a box of 121.63 km² that matches Asma's box (10.8 x 11.2 km, about 121 km², same centre), so the list stands. Full coloured table in the cloud coverage report.

Clear (23): 2022-11-04, 11-09, 11-14, 11-19, 11-24, 11-29, 12-04, 12-14, 12-19, 12-24, 12-29; 2023-01-03, 01-08, 01-13, 01-18, 01-23, 01-28, 02-02, 02-07, 02-22, 03-09, 03-24, 03-29

Partial (4): 2023-02-17, 2023-03-04, 2023-03-14, 2023-03-19

Cloudy (3): 2022-12-09, 2023-02-12, 2023-02-27

Use the clear dates; exclude partial and cloudy ones when computing vegetation indices.